

# Multiple Linear Regression

Computational Mathematics and Statistics

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# One Minute Paper Results

What was the most important thing you learned during this class?

What important question remains unanswered for you?

# Weight of Books

```
allbacks <- read.csv('../course_data/allbacks.csv')  
head(allbacks)
```

```
##      X volume area weight cover  
## 1 1      885  382    800    hb  
## 2 2     1016  468    950    hb  
## 3 3     1125  387   1050    hb  
## 4 4      239  371    350    hb  
## 5 5      701  371    750    hb  
## 6 6      641  367    600    hb
```

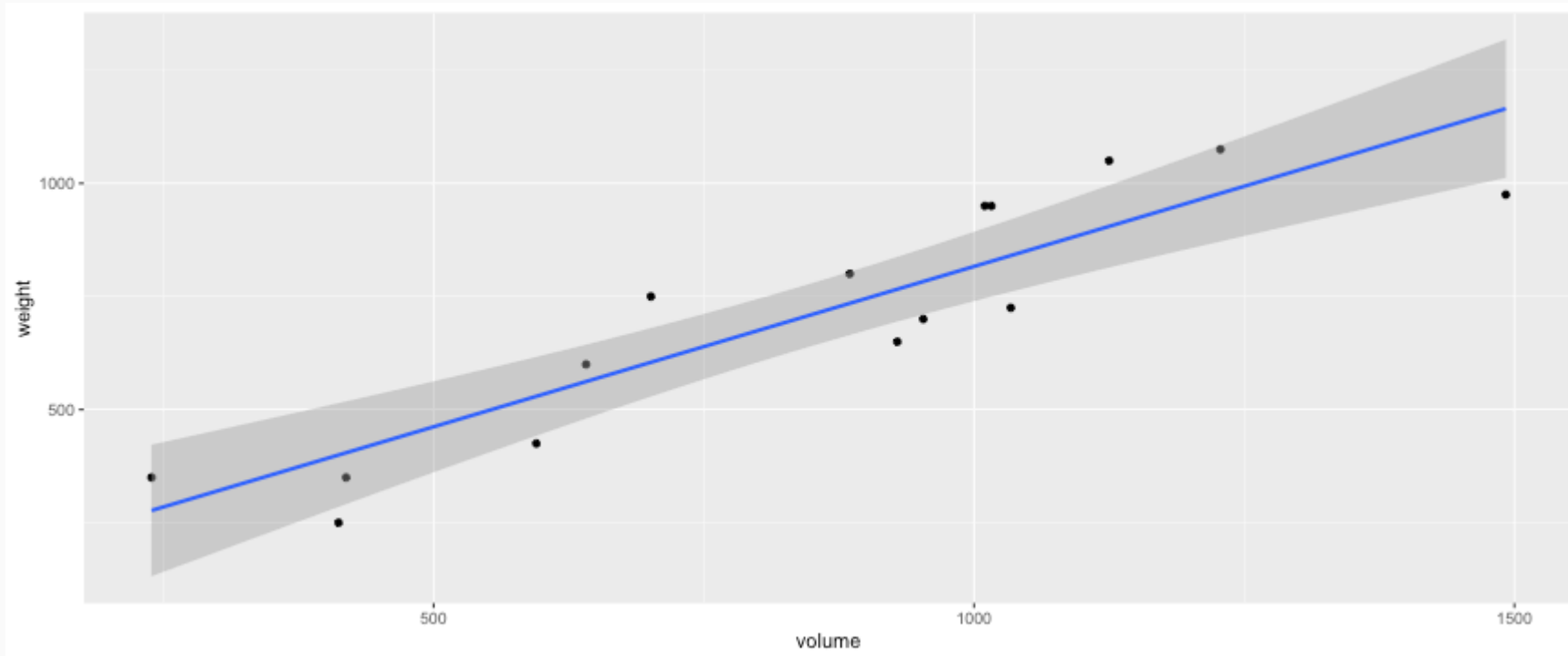
From: Maindonald, J.H. & Braun, W.J. (2007). *Data Analysis and Graphics Using R*, 2nd ed.

# Weights of Books (cont)

```
lm.out <- lm(weight ~ volume, data=allbacks)
```

$$\hat{weight} = 108 + 0.71volume$$

$$R^2 = 80\%$$



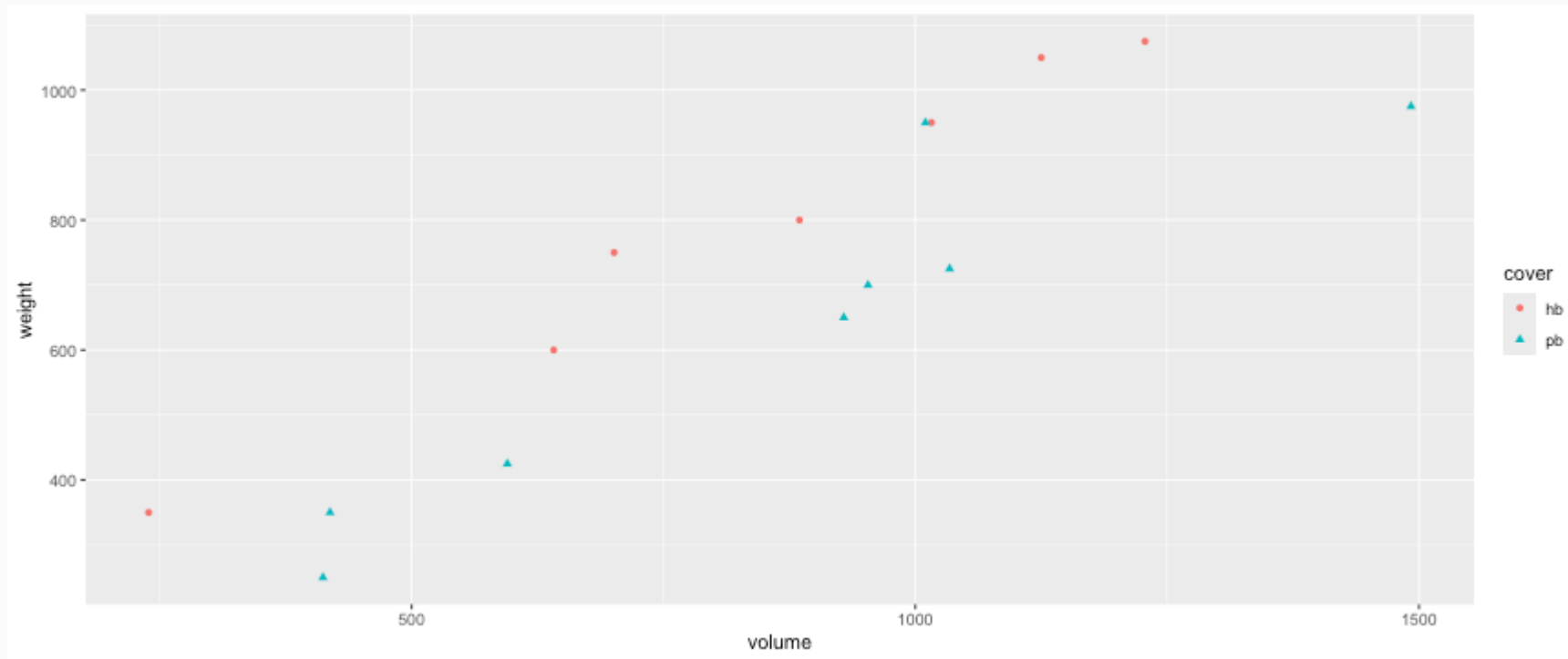
# Modeling weights of books using volume

```
summary(lm.out)
```

```
##
## Call:
## lm(formula = weight ~ volume, data = allbacks)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -189.97 -109.86   38.08  109.73  145.57
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 107.67931   88.37758   1.218   0.245
## volume       0.70864    0.09746   7.271 6.26e-06 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 123.9 on 13 degrees of freedom
## Multiple R-squared:  0.8026,    Adjusted R-squared:  0.7875
## F-statistic: 52.87 on 1 and 13 DF,  p-value: 6.262e-06
```

# Weights of hardcover and paperback books

- Can you identify a trend in the relationship between volume and weight of hardcover and paperback books?



- Paperbacks generally weigh less than hardcover books after controlling for book's volume.

# Modeling using volume and cover type

```
lm.out2 <- lm(weight ~ volume + cover, data=allbacks)
summary(lm.out2)
```

```
##
## Call:
## lm(formula = weight ~ volume + cover, data = allbacks)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -110.10  -32.32  -16.10   28.93  210.95
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  197.96284    59.19274   3.344 0.005841 **
## volume        0.71795     0.06153  11.669 6.6e-08 ***
## coverpb     -184.04727    40.49420  -4.545 0.000672 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 78.2 on 12 degrees of freedom
## Multiple R-squared:  0.9275,    Adjusted R-squared:  0.9154
## F-statistic: 76.73 on 2 and 12 DF,  p-value: 1.455e-07
```

# Linear Model

$$\hat{weight} = 198 + 0.72volume - 184coverpb$$

1. For **hardcover** books: plug in 0 for cover.

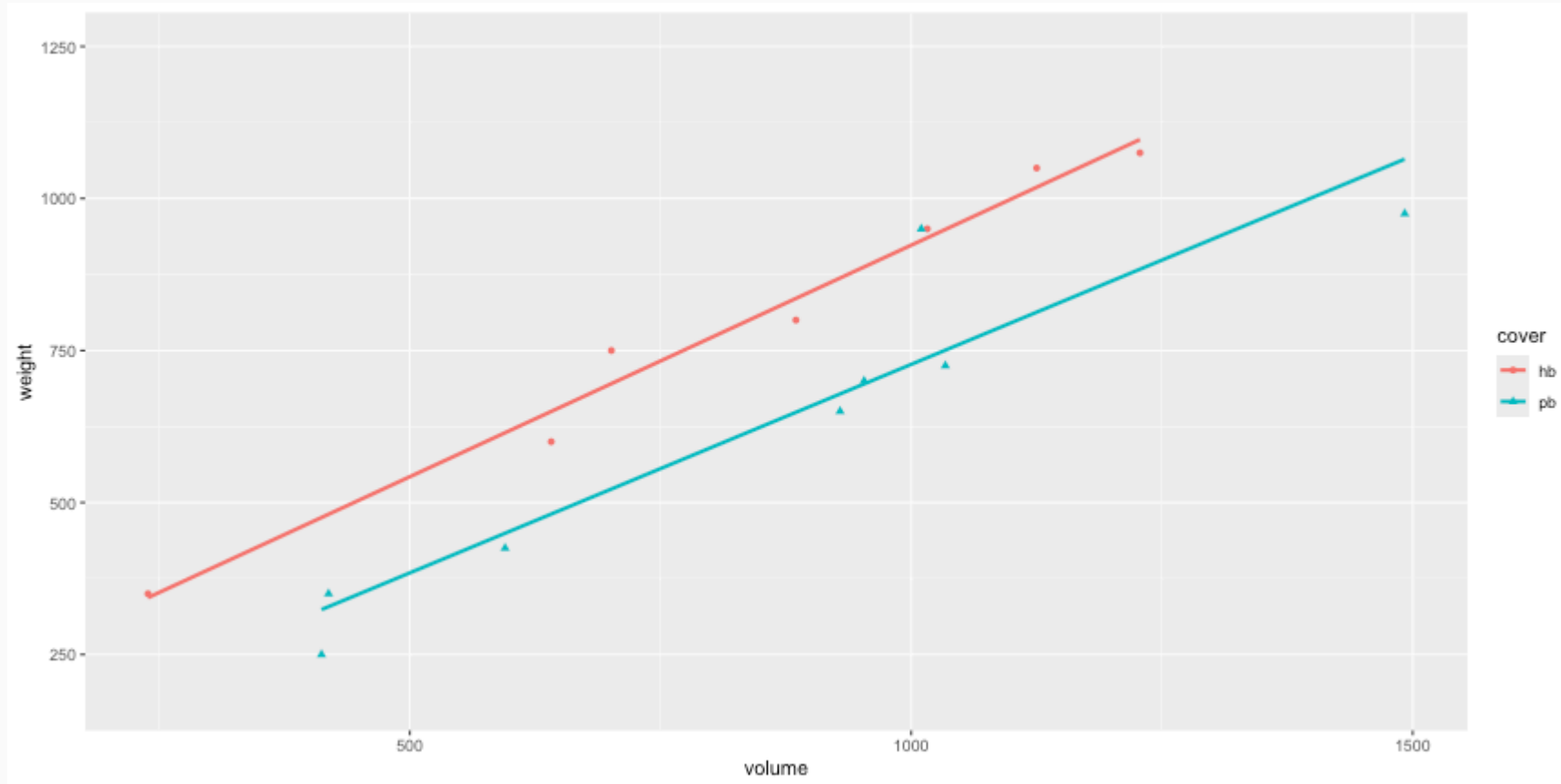
$$\hat{weight} = 197.96 + 0.72volume - 184.05 \times 0 = 197.96 + 0.72volume$$

1. For **paperback** books: put in 1 for cover.

$$\hat{weight} = 197.96 + 0.72volume - 184.05 \times 1$$



# Visualizing the linear model



# Interpretation of the regression coefficients

|             | Estimate  | Std. Error | t value | Pr(> t ) |
|-------------|-----------|------------|---------|----------|
| (Intercept) | 197.9628  | 59.1927    | 3.34    | 0.0058   |
| volume      | 0.7180    | 0.0615     | 11.67   | 0.0000   |
| coverpb     | -184.0473 | 40.4942    | -4.55   | 0.0007   |

- **Slope of volume:** All else held constant, books that are 1 more cubic centimeter in volume tend to weigh about 0.72 grams more.
- **Slope of cover:** All else held constant, the model predicts that paperback books weigh 184 grams lower than hardcover books.
- **Intercept:** Hardcover books with no volume are expected on average to weigh 198 grams.
  - Obviously, the intercept does not make sense in context. It only serves to adjust the height of the line.

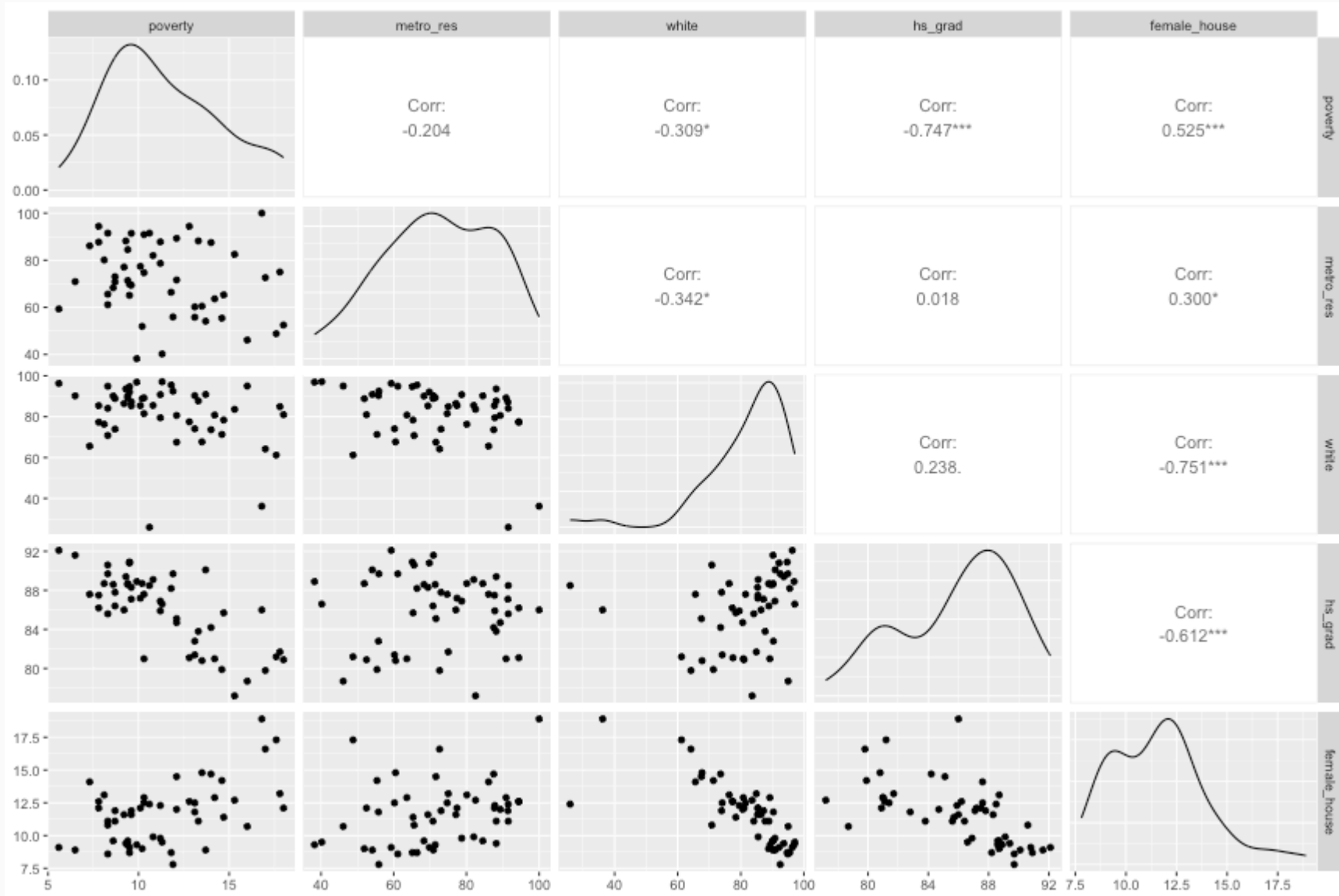
# Modeling Poverty

```
poverty <- read.table("../course_data/poverty.txt", h = T, sep = "\t")
names(poverty) <- c("state", "metro_res", "white", "hs_grad", "poverty", "female_house")
poverty <- poverty[,c(1,5,2,3,4,6)]
head(poverty)
```

| ##   | state      | poverty | metro_res | white | hs_grad | female_house |
|------|------------|---------|-----------|-------|---------|--------------|
| ## 1 | Alabama    | 14.6    | 55.4      | 71.3  | 79.9    | 14.2         |
| ## 2 | Alaska     | 8.3     | 65.6      | 70.8  | 90.6    | 10.8         |
| ## 3 | Arizona    | 13.3    | 88.2      | 87.7  | 83.8    | 11.1         |
| ## 4 | Arkansas   | 18.0    | 52.5      | 81.0  | 80.9    | 12.1         |
| ## 5 | California | 12.8    | 94.4      | 77.5  | 81.1    | 12.6         |
| ## 6 | Colorado   | 9.4     | 84.5      | 90.2  | 88.7    | 9.6          |

From: Gelman, H. (2007). *Data Analysis using Regression and Multilevel/Hierarchical Models*. Cambridge University Press.

# Modeling Poverty

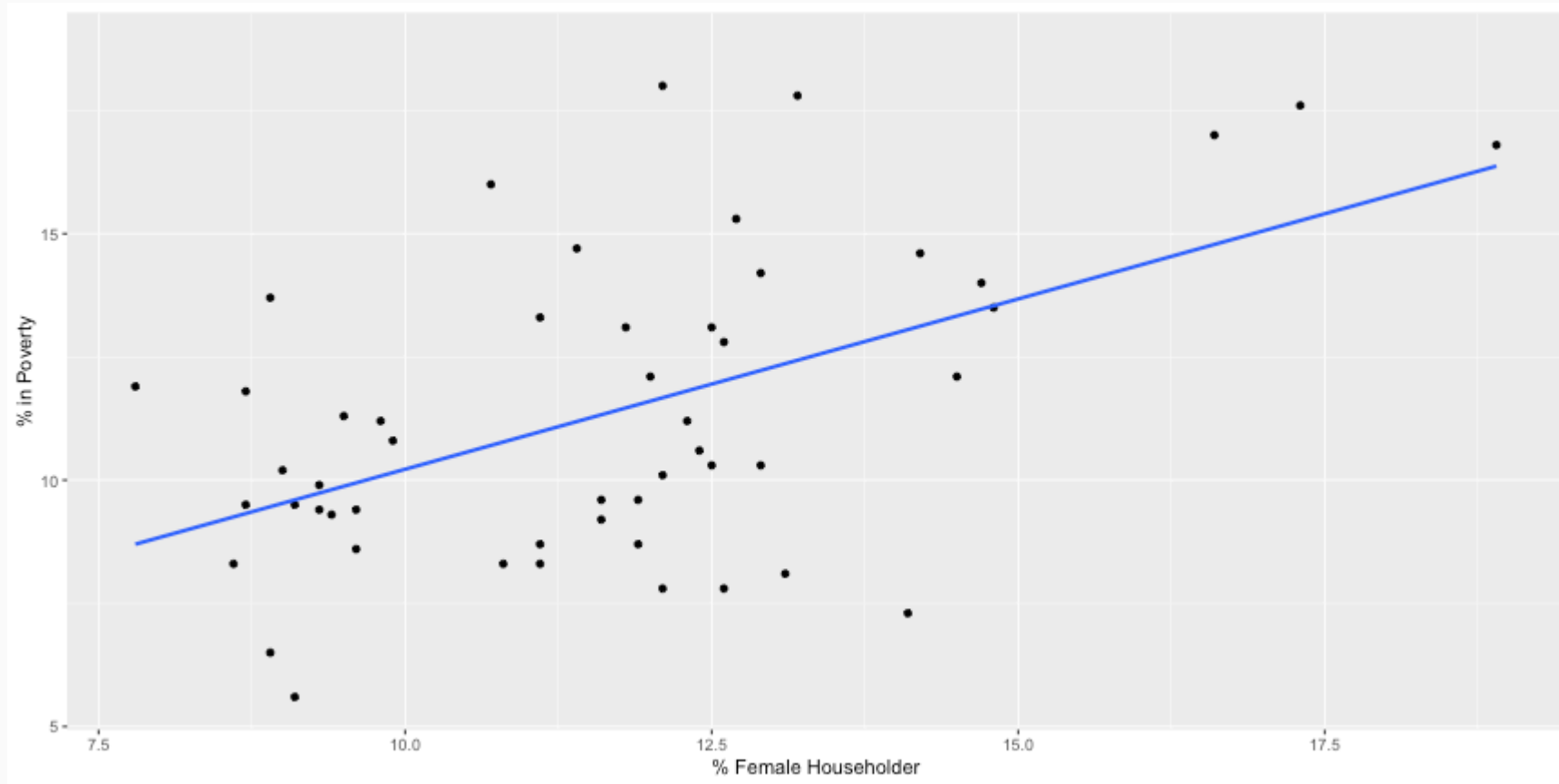


# Predicting Poverty using Percent Female Householder

```
lm.poverty <- lm(poverty ~ female_house, data=poverty)
summary(lm.poverty)
```

```
##
## Call:
## lm(formula = poverty ~ female_house, data = poverty)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -5.7537 -1.8252 -0.0375  1.5565  6.3285
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    3.3094     1.8970   1.745   0.0873 .
## female_house    0.6911     0.1599   4.322 7.53e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.664 on 49 degrees of freedom
## Multiple R-squared:  0.276,    Adjusted R-squared:  0.2613
## F-statistic: 18.68 on 1 and 49 DF,  p-value: 7.534e-05
```

# % Poverty by % Female Household



# Another look at $R^2$

$R^2$  can be calculated in three ways:

1. square the correlation coefficient of  $x$  and  $y$  (how we have been calculating it)
2. square the correlation coefficient of  $y$  and  $\hat{y}$
3. based on definition:

$$R^2 = \frac{\text{explained variability in } y}{\text{total variability in } y}$$

Using ANOVA we can calculate the explained variability and total variability in  $y$ .

# Sum of Squares

```
anova.poverty <- anova(lm.poverty)
print(xtable::xtable(anova.poverty, digits = 2), type='html')
```

|              | Df    | Sum Sq | Mean Sq | F value | Pr(>F) |
|--------------|-------|--------|---------|---------|--------|
| female_house | 1.00  | 132.57 | 132.57  | 18.68   | 0.00   |
| Residuals    | 49.00 | 347.68 | 7.10    |         |        |

```
sum(anova.poverty$`Sum Sq`)
```

```
## [1] 480.2475
```

```
sum((poverty$poverty - mean(poverty$poverty))^2)
```

```
## [1] 480.2475
```



# Sum of Squares (cont)

Sum of squares of  $y$ :  $SS_{Total} = \sum (y - \bar{y})^2 = 480.25 \rightarrow$  total variability

Sum of squares of residuals:  $SS_{Error} = \sum e_i^2 = 347.68 \rightarrow$  unexplained variability

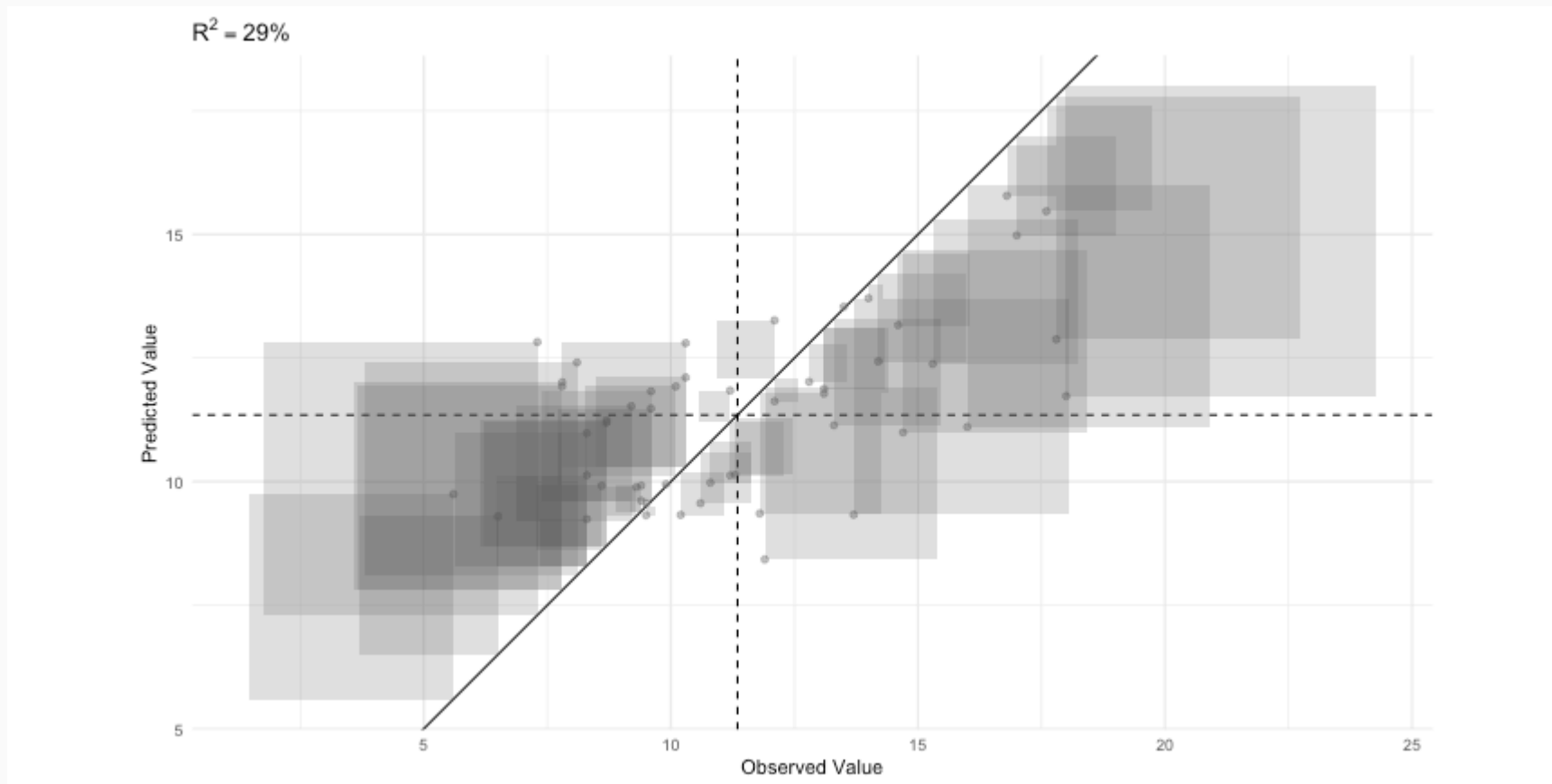
Sum of squares of  $x$ :  $SS_{Model} = SS_{Total} - SS_{Error} = 132.57 \rightarrow$  explained variability

$$R^2 = \frac{\text{explained variability in } y}{\text{total variability in } y} = \frac{132.57}{480.25} = 0.28$$

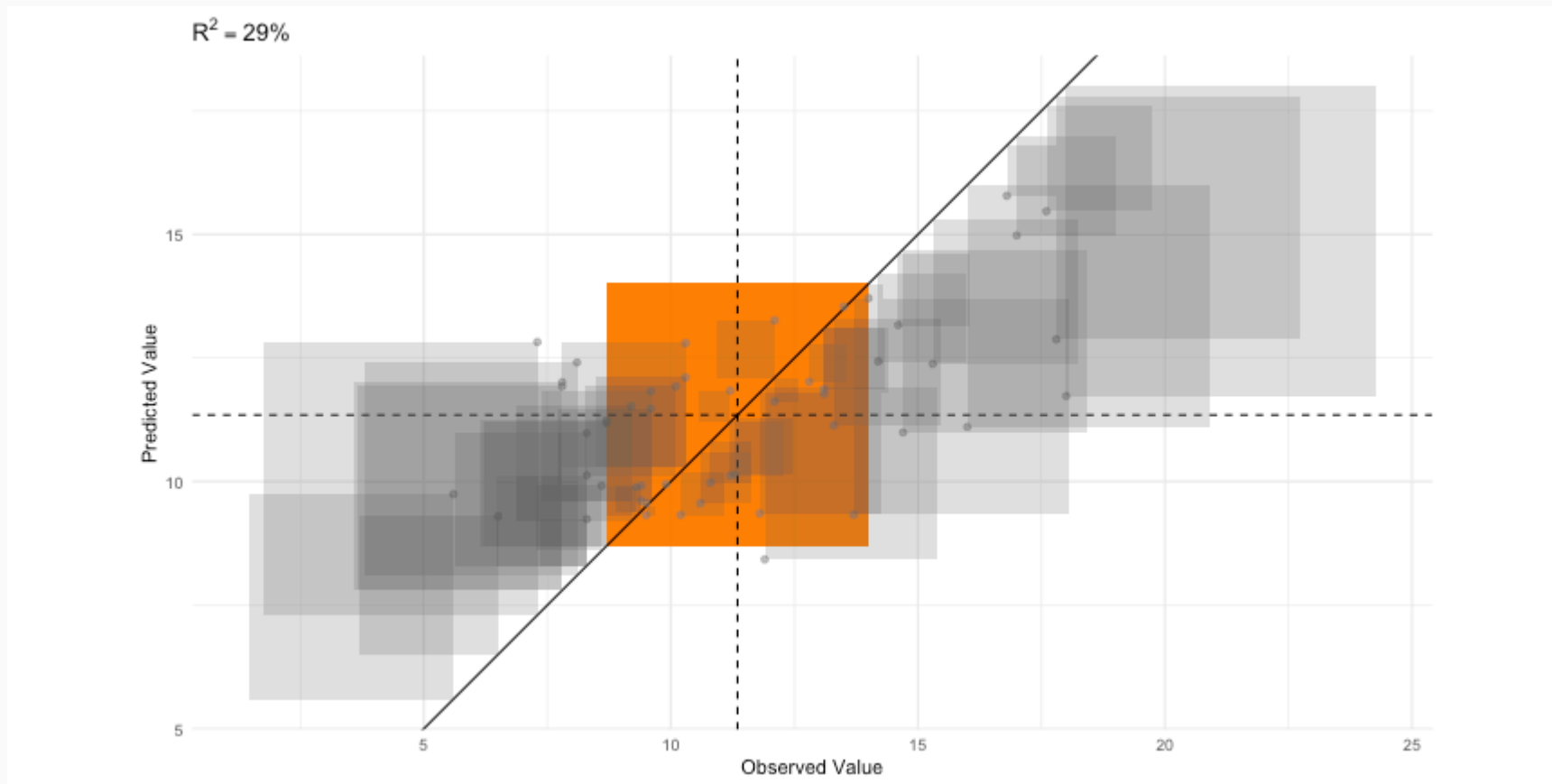
# Why bother?

- For single-predictor linear regression, having three ways to calculate the same value may seem like overkill.
- However, in multiple linear regression, we can't calculate  $R^2$  as the square of the correlation between  $x$  and  $y$  because we have multiple  $x$ s.
- And next we'll learn another measure of explained variability, *adjusted*  $R^2$ , that requires the use of the third approach, ratio of explained and unexplained variability.

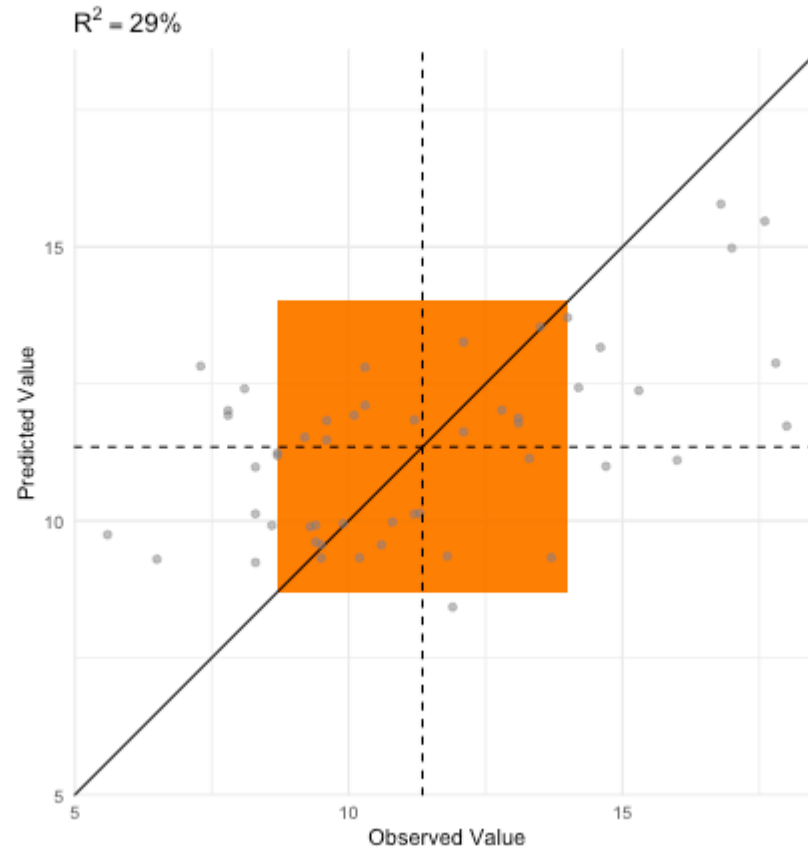
# $R^2$ : Error variance



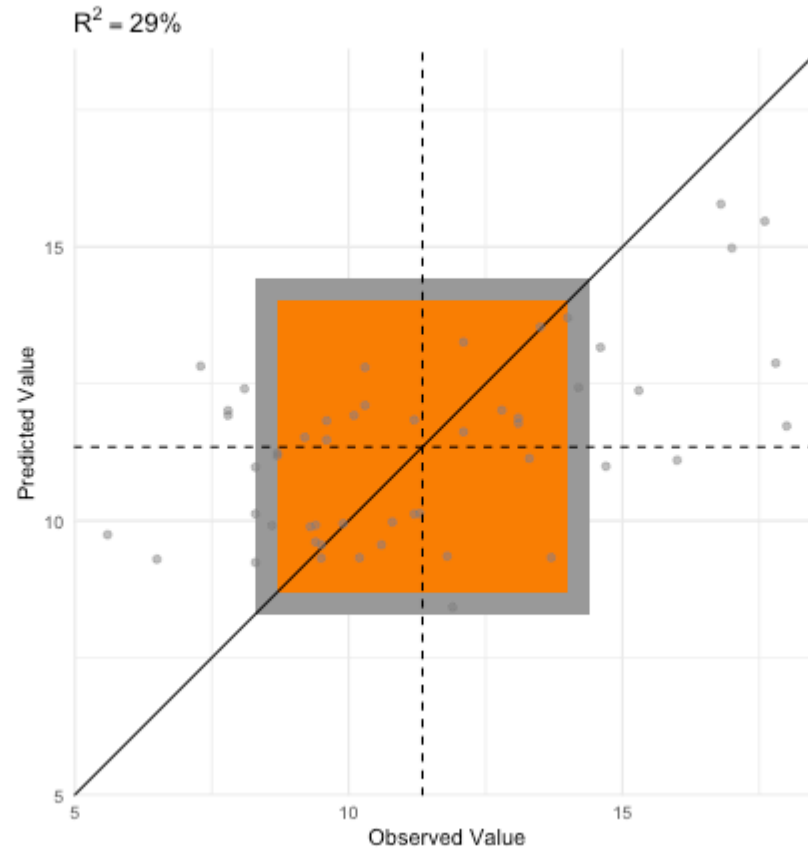
# $R^2$ : Error variance (cont.)



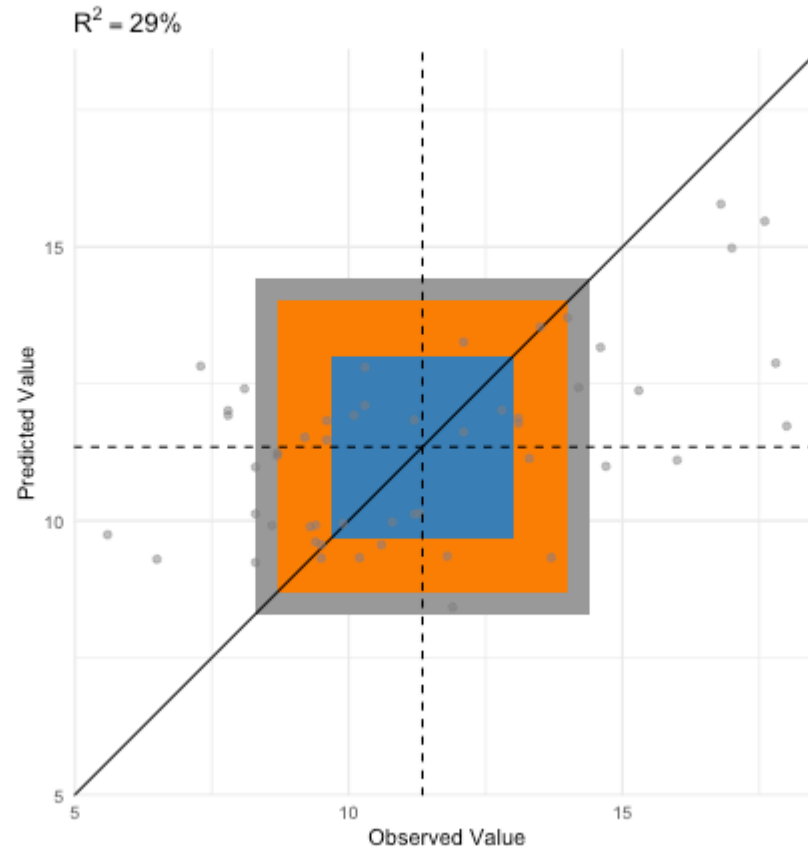
# $R^2$ : Error variance (cont.)



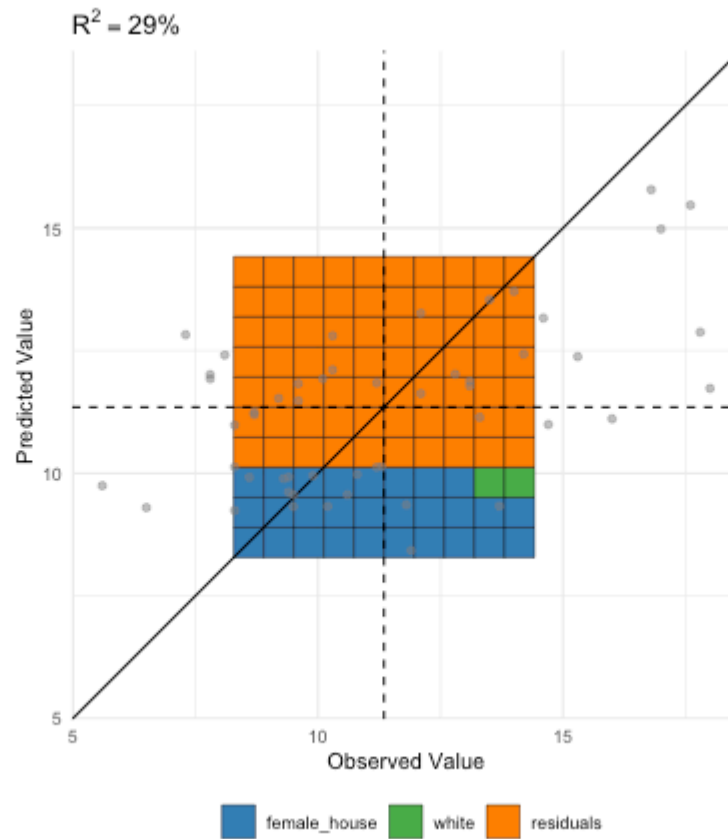
# $R^2$ : Total (grey) and error (orange) variance



# $R^2$ : Total (grey), error (orange), and regression (blue)



# $R^2$ : All variances



`VisualStats::r_squared_shiny()`



# Predicting poverty using % female household & %

```
lm.poverty2 <- lm(poverty ~ female_house + white, data=poverty)  
print(xtable::xtable(lm.poverty2), type='html')
```

|              | Estimate | Std. Error | t value | Pr(> t ) |
|--------------|----------|------------|---------|----------|
| (Intercept)  | -2.5789  | 5.7849     | -0.45   | 0.6577   |
| female_house | 0.8869   | 0.2419     | 3.67    | 0.0006   |
| white        | 0.0442   | 0.0410     | 1.08    | 0.2868   |

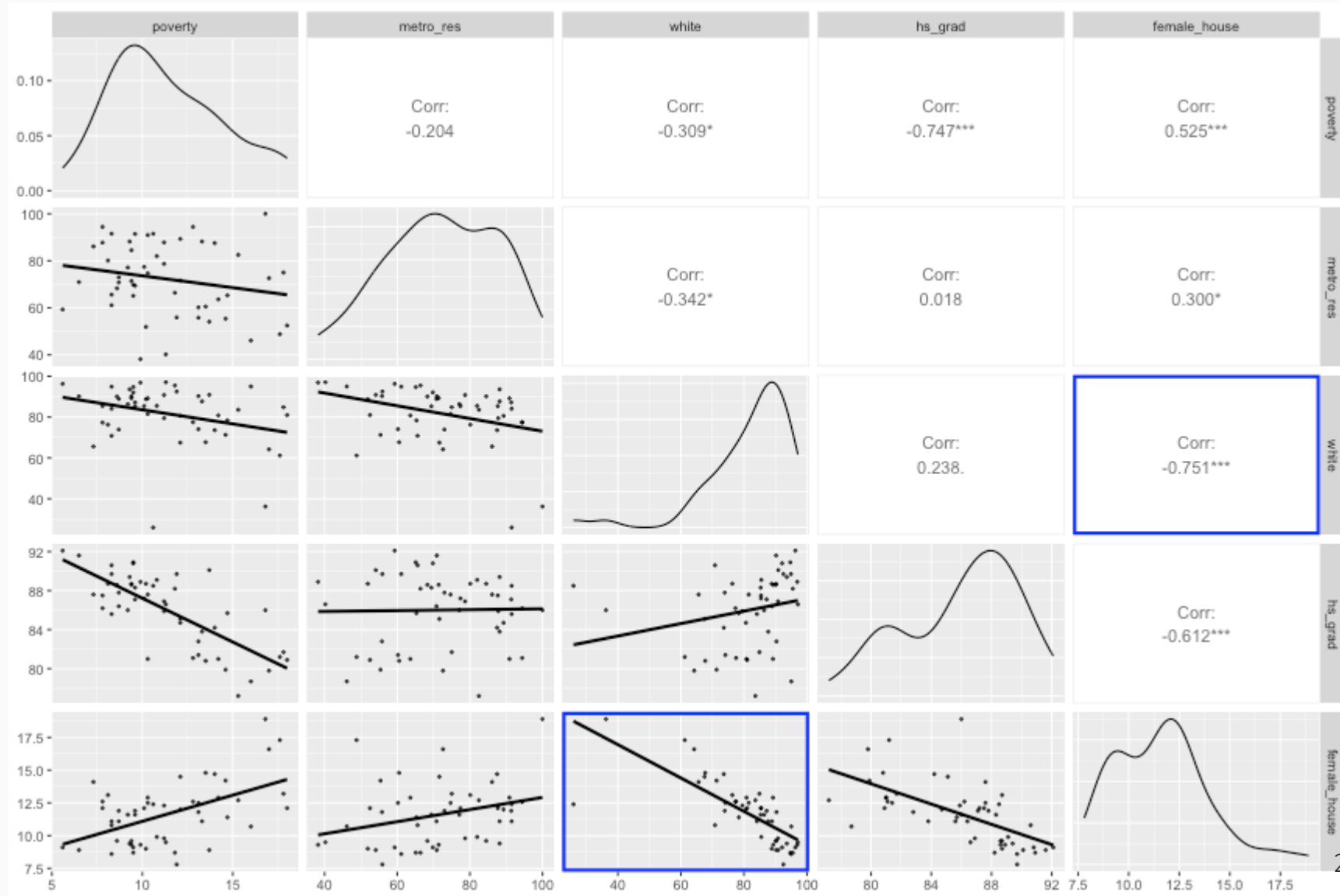
```
anova.poverty2 <- anova(lm.poverty2)  
print(xtable::xtable(anova.poverty2, digits = 3), type='html')
```

|              | Df     | Sum Sq  | Mean Sq | F value | Pr(>F) |
|--------------|--------|---------|---------|---------|--------|
| female_house | 1.000  | 132.568 | 132.568 | 18.745  | 0.000  |
| white        | 1.000  | 8.207   | 8.207   | 1.160   | 0.287  |
| Residuals    | 48.000 | 339.472 | 7.072   |         |        |

$$R^2 = \frac{\text{explained variability in } y}{\text{total variability in } y} = \frac{132.57 + 8.21}{480.25} = 0.29$$

# Unique information

Does adding the variable `white` to the model add valuable information that wasn't provided by `female_house`?



# Collinearity between explanatory variables

poverty vs % female head of household

|              | <b>Estimate</b> | <b>Std. Error</b> | <b>t value</b> | <b>Pr(&gt; t )</b> |
|--------------|-----------------|-------------------|----------------|--------------------|
| (Intercept)  | 3.3094          | 1.8970            | 1.74           | 0.0873             |
| female_house | 0.6911          | 0.1599            | 4.32           | 0.0001             |

poverty vs % female head of household and % female household

|              | <b>Estimate</b> | <b>Std. Error</b> | <b>t value</b> | <b>Pr(&gt; t )</b> |
|--------------|-----------------|-------------------|----------------|--------------------|
| (Intercept)  | -2.5789         | 5.7849            | -0.45          | 0.6577             |
| female_house | 0.8869          | 0.2419            | 3.67           | 0.0006             |
| white        | 0.0442          | 0.0410            | 1.08           | 0.2868             |

Note the difference in the estimate for `female_house`.

# Collinearity between explanatory variables

- Two predictor variables are said to be collinear when they are correlated, and this collinearity complicates model estimation.

Remember: Predictors are also called explanatory or independent variables. Ideally, they would be independent of each other.

- We don't like adding predictors that are associated with each other to the model, because often times the addition of such variable brings nothing to the table. Instead, we prefer the simplest best model, i.e. *parsimonious* model.
- While it's impossible to avoid collinearity from arising in observational data, experiments are usually designed to prevent correlation among predictors

# $R^2$ vs. adjusted $R^2$

| Model                      | $R^2$ | Adjusted $R^2$ |
|----------------------------|-------|----------------|
| Model 1 (Single-predictor) | 0.28  | 0.26           |
| Model 2 (Multiple)         | 0.29  | 0.26           |

- When any variable is added to the model  $R^2$  increases.
- But if the added variable doesn't really provide any new information, or is completely unrelated, adjusted  $R^2$  does not increase.

# Adjusted $R^2$

$$R^2_{adj} = 1 - \left( \frac{SS_{error}}{SS_{total}} \times \frac{n - 1}{n - p - 1} \right)$$

where  $n$  is the number of cases and  $p$  is the number of predictors (explanatory variables) in the model.

- Because  $p$  is never negative,  $R^2_{adj}$  will always be smaller than  $R^2$ .
- $R^2_{adj}$  applies a penalty for the number of predictors included in the model.
- Therefore, we choose models with higher  $R^2_{adj}$  over others.

# $R^2$ Three Ways

$R^2$  between mpg (miles per gallon) and wt (weight).

## 1. Squared correlation

```
data(mtcars)
cor(mtcars$mpg, mtcars$wt)^2
```

```
## [1] 0.7528328
```

## 2. Square of the correlation between $y$ and $\hat{y}$ .

```
lm_out <- lm(mpg ~ wt, data = mtcars)
mtcars$mpg_predicted <- predict(lm_out)
cor(mtcars$mpg, mtcars$mpg_predicted)^2
```

```
## [1] 0.7528328
```

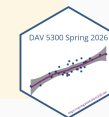
$$3. R^2 = \frac{\text{explained variability in } y}{\text{total variability in } y}$$

```
(anova_out <- anova(lm_out))
```

```
## Analysis of Variance Table
##
## Response: mpg
##           Df Sum Sq Mean Sq F value    Pr(>F)
## wt           1  847.73   847.73   91.375 1.294e-10 ***
## Residuals  30   278.32     9.28
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.'
```

```
total_ss <- sum((mtcars$mpg - mean(mtcars$mpg))^2)
sum(anova_out$`Sum Sq`[1]) / total_ss
```

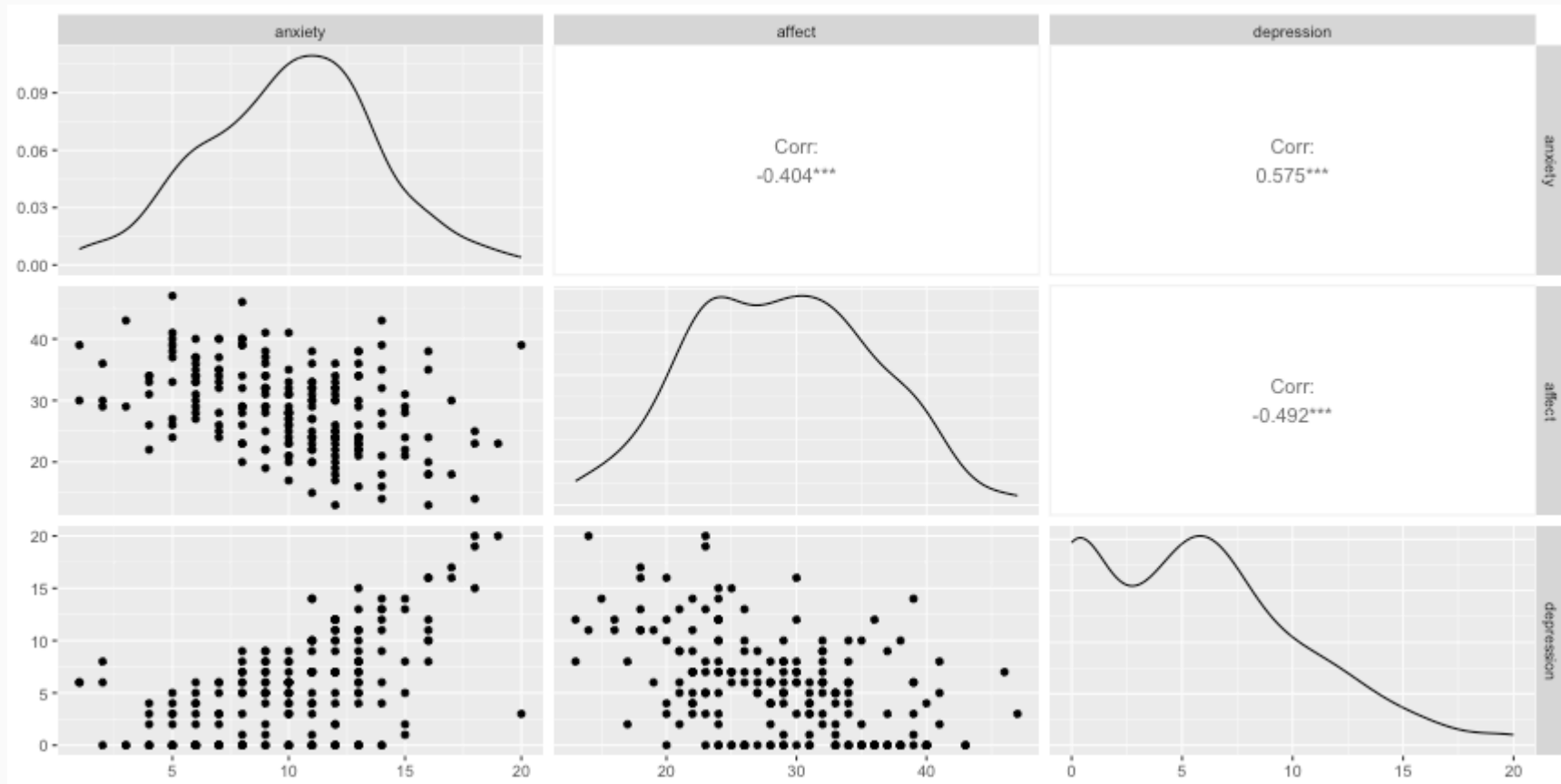
```
## [1] 0.7528328
```



# Interaction Effects

```
data("depression", package = "VisualStats")
```

```
GGally::ggpairs(depression)
```





# Interaction Effects (cont.)

```
lm(depression ~ anxiety + affect, data = depression) |> summary()
```

```
##
## Call:
## lm(formula = depression ~ anxiety + affect, data = depression)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -7.4298 -2.2581 -0.2697  2.6613  8.3104
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   5.73635    1.66764   3.440 0.000711 ***
## anxiety        0.58300    0.07755   7.518 1.92e-12 ***
## affect       -0.21048    0.04054  -5.191 5.18e-07 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.655 on 197 degrees of freedom
## Multiple R-squared:  0.4111,    Adjusted R-squared:  0.4051
## F-statistic: 68.77 on 2 and 197 DF,  p-value: < 2.2e-16
```

```
lm(depression ~ anxiety * affect, data = depression) |> summary()
```

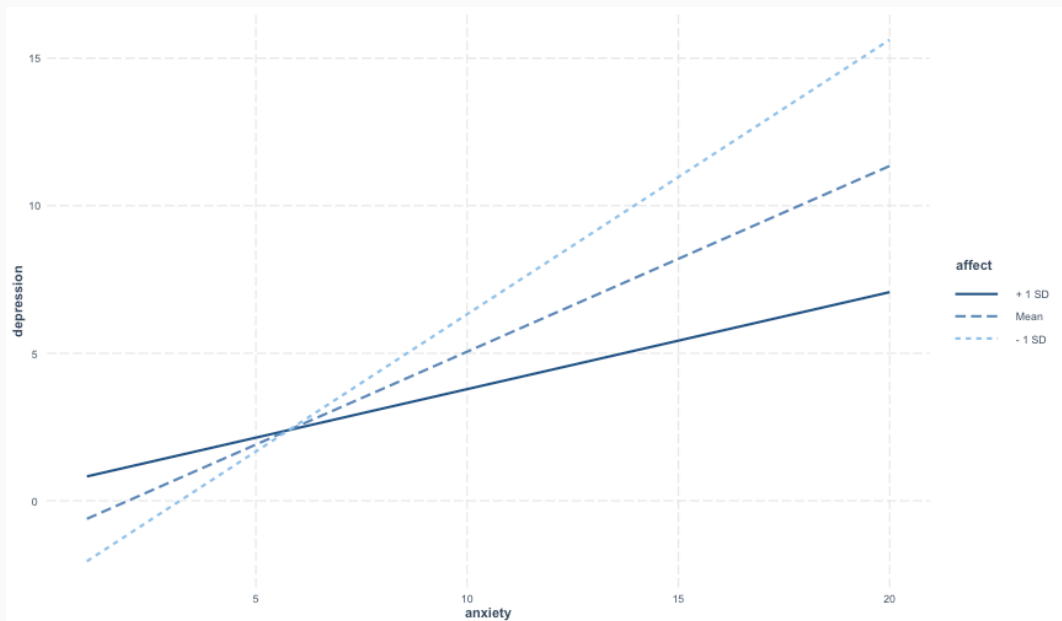
```
##
## Call:
## lm(formula = depression ~ anxiety * affect, data = depression)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -8.1177 -2.4875 -0.2649  2.3148 10.0265
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  -8.413732   3.530368  -2.383  0.0181 *
## anxiety        1.870338   0.296079   6.317 1.75e-09 ***
## affect         0.248703   0.109335   2.275  0.0240 *
## anxiety:affect -0.043035   0.009583  -4.491 1.21e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.489 on 196 degrees of freedom
## Multiple R-squared:  0.4661,    Adjusted R-squared:  0.4579
## F-statistic: 57.03 on 3 and 196 DF,  p-value: < 2.2e-16
```

# Visualizing Interaction Effects

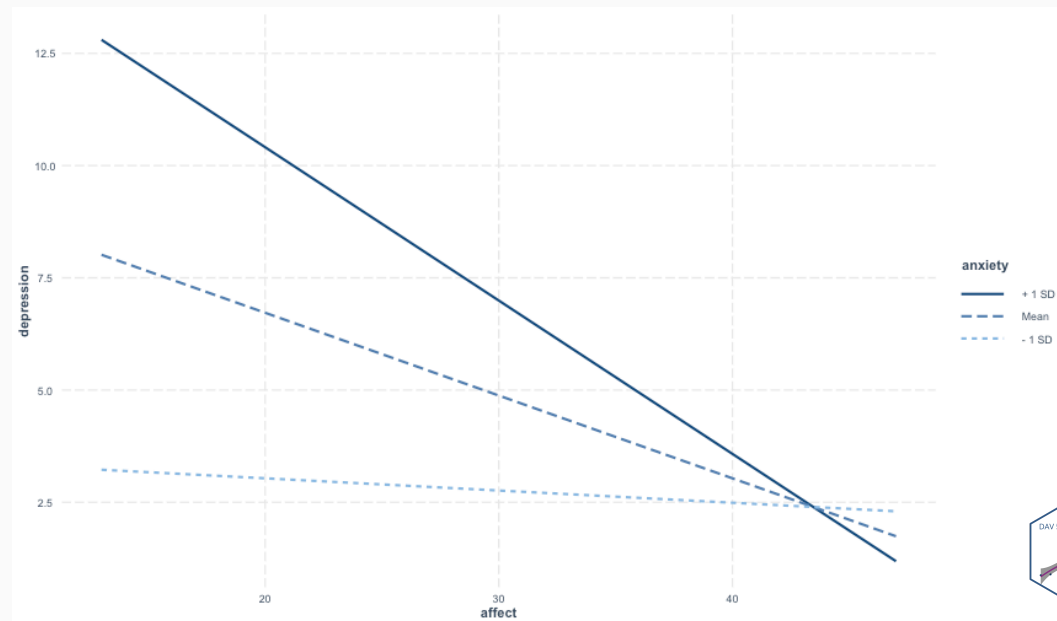
```
shiny::runApp(file.path(find.package('VisualStats'), 'shiny', 'multiple_regression'))
```

```
lm_out <- lm(depression ~ anxiety * affect, data = depression)
```

```
interactions::interact_plot(lm_out,  
  pred = anxiety, modx = affect)
```



```
interactions::interact_plot(lm_out,  
  pred = affect, modx = anxiety)
```



# ANOVA vs. Multiple Regression

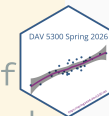
```
data("hand_washing", package = "VisualStats")
```

```
aov(Bacterial_Counts ~ Method, data = hand_washing) |>
```

```
##              Df Sum Sq Mean Sq F value    Pr(>F)
## Method          3   29882     9961   7.064 0.00111 **
## Residuals       28   39484     1410
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.'
```

```
lm(Bacterial_Counts ~ Method, data = hand_washing) |> s
```

```
##
## Call:
## lm(formula = Bacterial_Counts ~ Method, data = hand_
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -72.50 -20.88  -1.00   18.12  101.00
##
## Coefficients:
##              Estimate Std. Error t value
## (Intercept)       37.50      13.28   2.825
## MethodAntibacterial Soap  55.00      18.78   2.929
## MethodSoap          68.50      18.78   3.648
## MethodWater         79.50      18.78   4.234
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.'
##
## Residual standard error: 37.55 on 28 degrees of f
## Multiple R-squared:  0.4308,    Adjusted R-squared:
## F-statistic: 7.064 on 3 and 28 DF,  p-value: 0.00111
```



# Summary Statistics

```
VisualStats::describe_by(hand_washing$Bacterial_Counts, group = hand_washing$Method)
```

```
## # A tibble: 4 × 15
##   item group1      n mean    sd median trimmed  mad   min   max range   skew
##   <int> <chr>   <int> <dbl> <dbl> <dbl>   <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
## 1     1 1 Alcoho...    8  37.5  26.6  34.5   37.5  25.2    5    82    77  0.297
## 2     2 2 Antiba...    8  92.5  42.0  91.5   92.5  30.4   20   164   144 -0.0161
## 3     3 3 Soap      8 106    47.0 105    106    25.9   51   207   156  0.961
## 4     4 4 Water     8 117    31.1 114.    117    33.4   74   170    96  0.224
## # i 3 more variables: kurtosis <dbl>, se <dbl>, IQR <dbl>
```

# Dummy Coding

For regression, all independent variables must be quantitative. Therefore we must code categorical as a sequence of  $k - 1$  variables where the values are either 0 or 1. For the `hand_washing` dataset, one group is picked as the "reference group." The `lm` and `glm` functions will call the `model.matrix` automatically to do the dummy coding.

```
hand_washing |>  
  head()
```

| ##   | Bacterial_Counts | Method             |
|------|------------------|--------------------|
| ## 1 | 74               | Water              |
| ## 2 | 84               | Soap               |
| ## 3 | 70               | Antibacterial Soap |
| ## 4 | 51               | Alcohol Spray      |
| ## 5 | 135              | Water              |
| ## 6 | 51               | Soap               |

```
model.matrix(Bacterial_Counts ~ Method,  
             data = hand_washing)[,-1] |> head()
```

| ##   | MethodAntibacterial Soap | MethodSoap | MethodWater |
|------|--------------------------|------------|-------------|
| ## 1 | 0                        | 0          | 1           |
| ## 2 | 0                        | 1          | 0           |
| ## 3 | 1                        | 0          | 0           |
| ## 4 | 0                        | 0          | 0           |
| ## 5 | 0                        | 0          | 1           |
| ## 6 | 0                        | 1          | 0           |

# Defining your own reference group

Sometimes the reference group R picks may not be the one you want. For example, I tend to prefer the largest group to be the reference group.

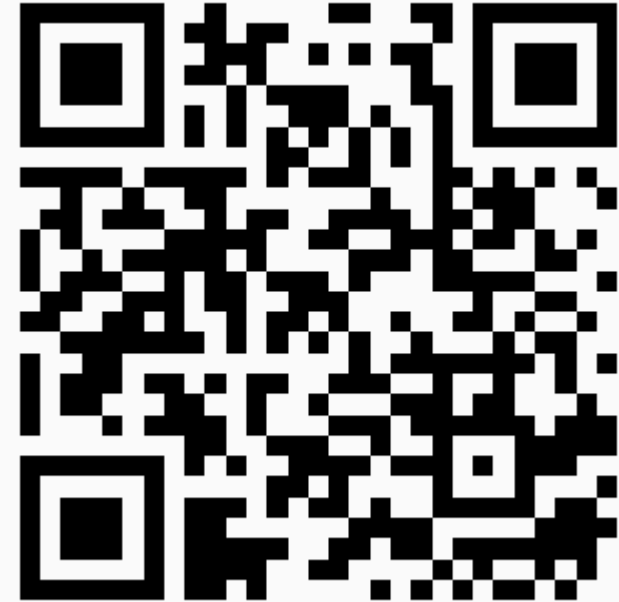
```
hand_washing <- hand_washing |> dplyr::mutate(Method2 = as.factor(Method)) |> dplyr::mutate(Method2 = relevel(Method2, ref = 'Water'))
```

```
lm(Bacterial_Counts ~ Method2, data = hand_washing) |> summary()
```

```
##
## Call:
## lm(formula = Bacterial_Counts ~ Method2, data = hand_washing)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -72.50 -20.88  -1.00   18.12  101.00
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)      117.00      13.28   8.813 1.45e-09 ***
## Method2Alcohol Spray      -79.50      18.78  -4.234 0.000224 ***
## Method2Antibacterial Soap  -24.50      18.78  -1.305 0.202563
## Method2Soap       -11.00      18.78  -0.586 0.562665
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 37.55 on 28 degrees of freedom
## Multiple R-squared:  0.4308,    Adjusted R-squared:  0.3698
## F-statistic: 7.064 on 3 and 28 DF,  p-value: 0.001111
```

# One Minute Paper

1. What was the most important thing you learned during this class?
2. What important question remains unanswered for you?



<https://forms.gle/hWUktVZ4Fyiia3xy6>